

2D Nodes Table Documentation

The **2D Nodes Table** is used to store the nodes for the **2D geometry** of the wing and fuselage. It contains 6 columns:

Column Name	Description	Example
local_id	Unique identifier for each node within its local part of the wing. It ensures traceability during the creation and export of nodes.	1, 2, 3, ...
x	xxx-coordinate of the node in the 2D plane.	0.25, 2.5, 5.0
y	yyy-coordinate of the node in the 2D plane.	0.0, 0.5, 1.2
rib_index	Index of the rib where the node is located. This distinguishes the rib-wise position of the node in the wing structure.	1, 2, 3
stringer_index	Index of the stringer where the node is located. This distinguishes the stringer-wise position of the node in the wing structure.	1, 2, 3
tag	Tag used to identify the structural part of the node. The tag distinguishes between different parts of the wing and fuselage geometry.	See Tag Definitions below

Tag Definitions for 2D Nodes

The tag column identifies the structural context of each node and is crucial for organizing the data. Below are the definitions for each tag:

Tag	Description
rear spar	Node located along the rear spar of the wing.
front spar	Node located along the front spar of the wing.
stringer	Node located along the stringers of the wing.
rear spar fuselage	Node located along the rear spar of the fuselage (at the wing-fuselage intersection).
front spar fuselage	Node located along the front spar of the fuselage (at the wing-fuselage intersection).

Tag	Description
stringer fuselage	Node located along the stringers of the fuselage (at the wing-fuselage intersection).
OnlyNode	A critical node at the corner of the root-front_spar_wing-front_spar_fuselage. It serves as a unique reference point.

3D Nodes Table Documentation

The **3D Nodes Table** represents the 3D geometry of the wing and fuselage, adding a third dimension (zzz) and distinguishing between **extrados** (top) and **intrados** (bottom) surfaces. It contains 8 columns:

Column Name	Description	Example
local_id	Unique identifier for each node within its local part of the wing. It is inherited from the 2D node table and ensures traceability during the 2D-to-3D transition.	1, 2, 3, ...
x	xxx-coordinate of the node in the 3D space.	0.25, 2.5, 5.0
y	yyy-coordinate of the node in the 3D space.	0.0, 0.5, 1.2
z	zzz-coordinate of the node in the 3D space. The zzz-coordinate distinguishes between the extrados ($z=H/2$ $z=H/2z=H/2$) and intrados ($z=-H/2$ $z=-H/2z=-H/2$).	0.05, -0.05
rib_index	Index of the rib where the node is located. This is inherited from the 2D table and helps organize the rib-wise position of the node.	1, 2, 3
stringer_index	Index of the stringer where the node is located. This is inherited from the 2D table and helps organize the stringer-wise position of the node.	1, 2, 3
tag	Tag used to identify the structural part of the node, inherited from the 2D table.	See Tag Definitions below

Column Name	Description	Example
h	Indicates whether the node belongs to the extrados or intrados surface.	extrados, intrados

Tag Definitions for 3D Nodes

The tag column in the 3D table inherits the values from the 2D table but now includes information about the extrados and intrados surfaces. Below are the definitions:

Tag	Description
rear spar	Node located along the rear spar of the wing.
front spar	Node located along the front spar of the wing.
stringer	Node located along the stringers of the wing.
rear spar fuselage	Node located along the rear spar of the fuselage (at the wing-fuselage intersection).
front spar fuselage	Node located along the front spar of the fuselage (at the wing-fuselage intersection).
stringer fuselage	Node located along the stringers of the fuselage (at the wing-fuselage intersection).
OnlyNode	A critical node at the corner of the root-front_spar_wing-front_spar_fuselage. It serves as a unique reference point.

2D-to-3D Transition Process

To generate the **3D Nodes Table** from the **2D Nodes Table**, the following steps are applied:

- Input:** Start with the 2D Nodes Table (6 columns: local_id, x, y, rib_index, stringer_index, tag).
- Add zzz-Coordinate:**
 - For **extrados nodes**: $z = H/2$
 - For **intrados nodes**: $z = -H/2$
- Add h Column:**

- Assign "extrados" or "intrados" based on the zzz-coordinate.

4. Combine Nodes:

- Each 2D node is transformed into two 3D nodes: one for the extrados and one for the intrados.

5. Output: Generate the 3D Nodes Table (8 columns: local_id, x, y, z, rib_index, stringer_index, tag, h).

General Notes

- The OnlyNode tag is critical for ensuring proper alignment between the wing and fuselage. It should be carefully monitored during the 2D-to-3D transition.
 - The zzz-coordinate system uses a centered approach:
 - $z=H/2$ for the **extrados** (top surface).
 - $z=-H/2$ for the **intrados** (bottom surface).
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Example: 2D and 3D Node Tables

2D Nodes Table Example

local_id	x	y	rib_index	stringer_index	tag
1	0.0	0.0	1	1	front spar
2	1.0	0.0	1	2	stringer
3	0.5	0.1	1	-	OnlyNode

3D Nodes Table Example

local_id	x	y	z	rib_index	stringer_index	tag	h
1	0.0	0.0	0.05	1	1	front spar	extrados
1	0.0	0.0	-0.05	1	1	front spar	intrados
2	1.0	0.0	0.05	1	2	stringer	extrados
2	1.0	0.0	-0.05	1	2	stringer	intrados
3	0.5	0.1	0.05	1	-	OnlyNode	extrados

local_id	x	y	z	rib_index	stringer_index	tag	h
3	0.5	0.1	-0.05	1	-	OnlyNode	intrados